# Mandi Saathi — Startup & Execution Manual

AI Price & Sell-Timing Advisor for Indian Farmers • Official Developer & Operator Guide

## 1. Quick All-In-One Copy-Paste Commands

For instant execution, copy and paste either of the following single-line commands into your Windows PowerShell or CMD terminal:

| | |
|---|---|
| **Option A: Daily Instant Start**<br>Navigates to project directory and boots Uvicorn with auto-reload | `cd c:\Users\ajink\OneDrive\Desktop\project\mandisathi\mandiSaaathi; python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload` |
| **Option B: Full Setup & Start**<br>Installs/verifies all dependencies first, then boots the server | `cd c:\Users\ajink\OneDrive\Desktop\project\mandisathi\mandiSaaathi; pip install -r requirements.txt; python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload` |

## 2. Detailed Command Breakdown & What Each Command Does

Below is an in-depth breakdown of what happens behind the scenes for every individual command:

| Step & Command | What It Does & Why It Is Needed |
|---|---|
| **1. Change Directory**<br>`cd ...\mandiSaaathi` | Switches your active terminal session to the project directory root. This ensures all relative directory references like backend/, frontend/, and data/ resolve accurately. |
| **2. Create Virtual Env**<br>`python -m venv .venv` | Initializes a dedicated, isolated Python virtual environment inside .venv/. Prevents system package conflicts and isolates versions from other projects on your machine. |
| **3. Activate Virtual Env**<br>`.\.venv\Scripts\Activate.ps1`<br>*(or activate.bat in CMD)* | Sets your shell PATH to prioritize the virtual environment. Any subsequent pip install or python command operates strictly within this isolated sandbox. |
| **4. Install Packages**<br>`pip install -r requirements.txt` | Reads requirements.txt and installs core runtime dependencies: **FastAPI** (API framework), **Uvicorn** (ASGI web server), **Pandas & NumPy** (data engines), **Pydantic** (schema validation), and **HTTPX** (Gemini client). |
| **5. Environment Config**<br>`cp .env.example .env` | Creates your local configuration file from the sample template. You can optionally set GEMINI_API_KEY for AI natural language chat parsing, or custom PORT and HOST values. |
| **6. Launch Web Server**<br>`python -m uvicorn backend.main:app --host 127.0.0.1 --port 8000 --reload` | Boots the ASGI application on port 8000. **Crucial feature:** FastAPI automatically serves **both** the backend API and the static frontend UI simultaneously at http://localhost:8000 via Starlette static file mounting! |

## 3. Understanding the Uvicorn Command Arguments

| Argument / Flag | Technical Purpose & Function |
|---|---|
| `backend.main:app` | Tells Uvicorn to import module backend.main and locate the ASGI instance named app. |
| `--host 127.0.0.1` | Binds locally to localhost. Use --host 0.0.0.0 when deploying inside Docker or cloud containers (Render/Railway). |
| `--port 8000` | Sets the TCP port listener. If port 8000 is occupied, you can change this to 8080 or any free port. |
| `--reload` | Enables hot-reload mode for development. The server monitors Python files and reloads instantly when files change. |

## 4. Application Access URLs

| View / Resource | URL | Description |
|---|---|---|
| **Main App UI** | `http://localhost:8000` | Interactive Single Page App (Advisor, Chat, Alerts, Proof, About) |
| **Health Check** | `http://localhost:8000/api/health` | Returns {"status":"ok"} confirming backend is healthy |
| **Data Status** | `http://localhost:8000/api/data-status` | Displays status of 6,630 verified Agmarknet historical records |
| **Swagger Docs** | `http://localhost:8000/docs` | Interactive OpenAPI/Swagger documentation for testing all routes |

## 5. Utility & Testing Commands

| Action | Command | Description |
|---|---|---|
| **Run Tests** | `python -m pytest` | Executes all 51 automated test cases across the entire suite |
| **Backfill Data** | `python scripts/backfill.py` | Ingests 90 days of Agmarknet historical records into SQLite |
| **Load CSV** | `python scripts/load_manual_csv.py <file>` | Loads downloaded Agmarknet CSV into the SQLite database |
| **Stop Server** | `Press Ctrl + C` | Sends SIGINT to safely terminate the running Uvicorn server |

## 6. Common Troubleshooting Scenarios

| Issue / Error Message | Root Cause & Instant Resolution |
|---|---|
| `Unable to Connect to Mandi Engine` | The backend Uvicorn process is not running. Launch the server using python -m uvicorn backend.main:app --port 8000 and refresh the page. |
| `Port 8000 Already in Use (WinError 10048)` | Another background process is holding port 8000. Run netstat -ano \| findstr :8000 to find the PID and kill it, or launch on --port 8080. |
| `PowerShell ExecutionPolicy Error` | Run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass before activating the virtual environment. |